

Mini-project CAM C301 Week 4 & 5: Applying NLP for Topic Modelling

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2025-07-27

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1 Introduction

In this project, we will explore the application of automated topic modelling techniques for analyzing customer review data from PureGym. Topic modelling is crucial for business data analysis as it enables companies to identify key themes and patterns within large volumes of textual data, allowing for efficient extraction of insights without manual document analysis. We will apply Natural Language Processing (NLP) techniques including topic modelling using methods such as Latent Dirichlet Allocation (LDA) and emotion analysis using BERT (Bidirectional Encoder Representations from Transformers). These approaches are particularly effective for understanding customer sentiment and identifying emerging trends in unstructured text data. Finally, we will perform topic analysis and obtain actionable insights using a conversational large language model (LLM) such as microsoft/Phi-4-mini-instruct.

2 Objectives

- Use two data sets containing customer reviews from Google and Trustpilot to perform sentiment analysis.
- Using LDA and BERT-based emotion analysis, we'll identify key customer themes and sentiment patterns while testing how different preprocessing techniques affect model performance.
- The project will demonstrate how NLP techniques extract actionable business insights from unstructured text data. We will also experience the impact of feature engineering and hyperparameter tuning on topic coherence and sentiment accuracy.
- Our final report will summarize observations and performance differences between methodologies across both review platforms.

3 Methods

All preprocessing and initial analysis was implemented in Python 3.9.23 using Notebook 1 run locally. Topic analysis using the LLM model (microsoft/Phi-4-mini-instruct) was implemented in Notebook 2 and executed on Google Colab

server using T4 GPU.

Two real-world datasets containing 12 months of customer reviews were analyzed: Google Reviews (23,250 rows) and Trustpilot (16,673 rows). Datasets underwent preprocessing using pandas (v2.2.3). Irrelevant columns and rows with missing reviews were dropped, while rows with missing locations were identified as 'unknown'. Only reviews concerning UK-based locations were retained for analysis.

Text preprocessing involved removing numbers, English stop words, internet addresses, and symbols using NLTK tokenization. Words shorter than 3 letters and the terms 'pure' and 'gym' were removed. Word frequency analysis was conducted using nltk.FreqDist and WordCloud functions.

Negative reviews (score < 3) were selected from both datasets to reduce computational time. The datasets were subset based on common locations between platforms, then reduced to the top 30 locations with highest negative review counts. Topic modeling was performed using BERTopic and LDA, while emotion analysis utilized "bhadrash-savani/bert-base-uncased-emotion". Finally, topic analysis and extraction of actionable insights was performed with LLM microsoft/Phi-4-mini-instruct.

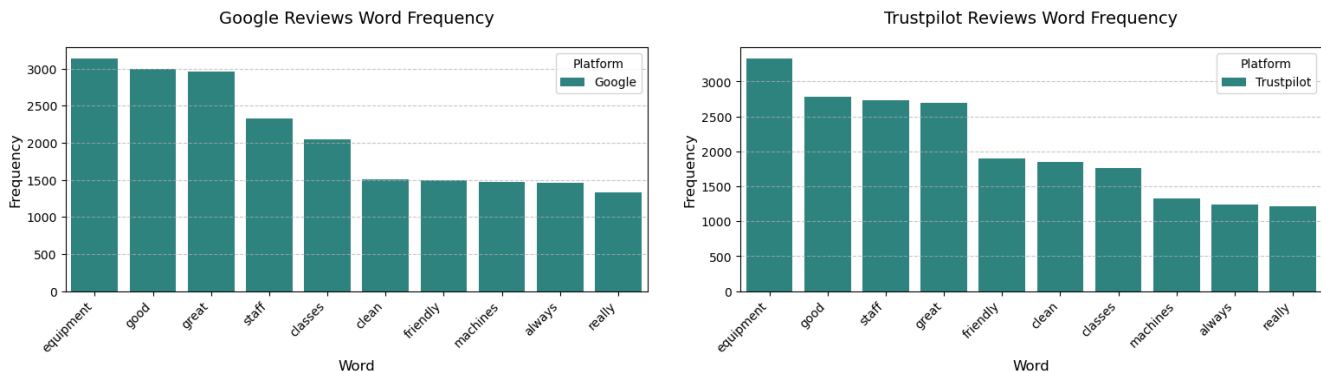


Figure 1: Word Frequency Analysis for All Reviews in UK Locations from Google Reviews and Trustpilot: Word frequency analysis of customer reviews collected from PureGym locations across the UK, using data sourced from both Google Reviews and Trustpilot. The figure visualizes the most common words appearing in customer feedback after text preprocessing steps such as the removal of stop words, numbers, symbols, and gym-specific terms (e.g., "pure", "gym"). The analysis encompasses both positive and negative reviews, providing a holistic view of the language most frequently used by customers.

4 Results

Following data preprocessing and word frequency analysis, the top 10 most frequent words showed a prevalence of positive terms (great, clean, good, friendly), aligning with 70.81% and 68.53% of reviews scoring 4-5 stars on Google and Trustpilot respectively. However, neutral descriptors like “equipment” and “staff” among frequent terms highlighted limitations of this basic frequency analysis in capturing contextual sentiment (See figure 1 and notebook 1).

This finding motivated implementation of advanced NLP techniques including topic modeling and emotion classification. To better understand negative sentiment while optimizing computing resources, subsequent analysis focused on reviews scoring below 3 from locations present in both datasets. Figure 2 provides a visual representation of where these reviews were most concentrated, highlighting potential areas for operational improvement.

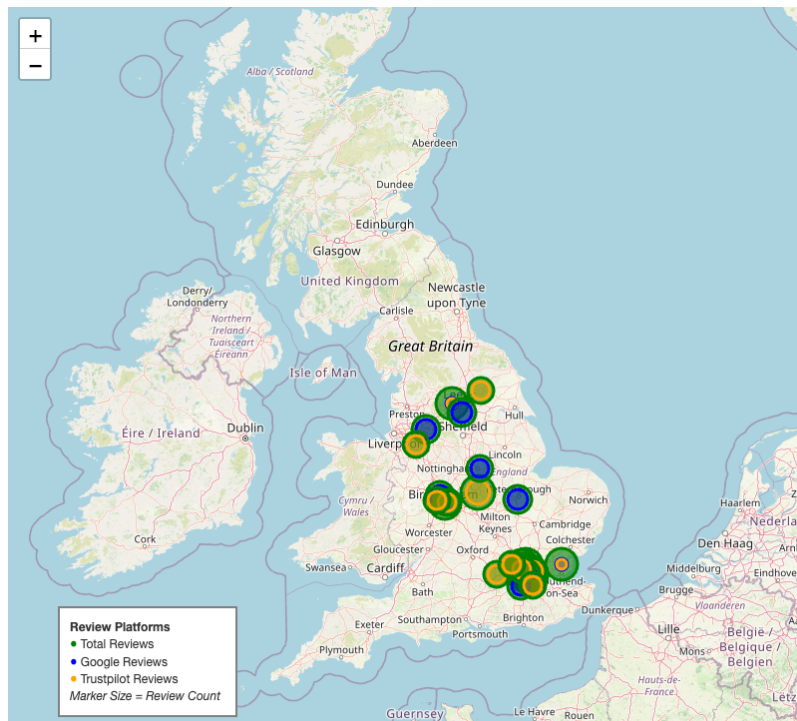


Figure 2: Distribution of Top 30 Locations with the Highest Negative Review Counts from Google Reviews and Trustpilot: The figure shows the distribution of the locations with the highest counts of negative reviews. Following the rubric instructions, the top 20 locations with the highest number of negative reviews were selected for each dataset. Next, these lists were concatenated and the top 30 locations with the highest number of negative reviews were selected. The size of each symbol corresponds to the number of negative reviews at that location, highlighting geographic hotspots of customer dissatisfaction.

Following text preprocessing, negative customer reviews underwent topic modeling using BERTopic to extract dominant thematic patterns. The analysis focused on examining topic distributions between all negative reviews from shared locations versus reviews from the top 30 locations with highest complaint volumes. Word cloud visualization revealed consistent high frequency of terms such as “equipment,” “clean,” “staff,” “weights,” and “machines,” indicating systematic concerns regarding facility maintenance and service delivery. Importantly, minimal differences were observed between the two subsets, suggesting that negative sentiment drivers are globally distributed rather than location-specific phenomena (Figure 3). This observation was confirmed by frequency distribution analysis, which revealed that equipment-related, cleanliness, and staff-related terms appeared amongst the most frequent terms used in both negative reviews subsets (Figure 4).



Figure 3: Comparison of the Most Frequent Topics Themes Among the top 30 Gyms with the Highest Negative Reviews versus All Negative Reviews from Shared Locations: Word cloud derived from negative customer reviews at the top 30 PureGym locations with the highest complaint volumes, combining data from both Google Reviews and Trustpilot. The word cloud was generated after rigorous text preprocessing, including removal of stop words, numbers, gym-specific terms (“pure”, “gym”), and short words, to isolate meaningful terms relevant to customer dissatisfaction.

Analysis of emotion distribution across negative reviews from the 30 highest-complaint PureGym locations using a BERT-based classifier revealed predominant emotions of anger and sadness (Figure 5). However, joy emerged as one of the most prevalent emotions, which contradicts expectations for negative reviews and suggests potential model misclassification or need for fine-tuning. The presence of anger and sadness indicates that customer dissatisfaction extends beyond logistical complaints, reflecting significant emotional impact from operational failures. These findings highlight the importance of model optimization before drawing conclusions about the relationship between complaint

themes and customer sentiment.

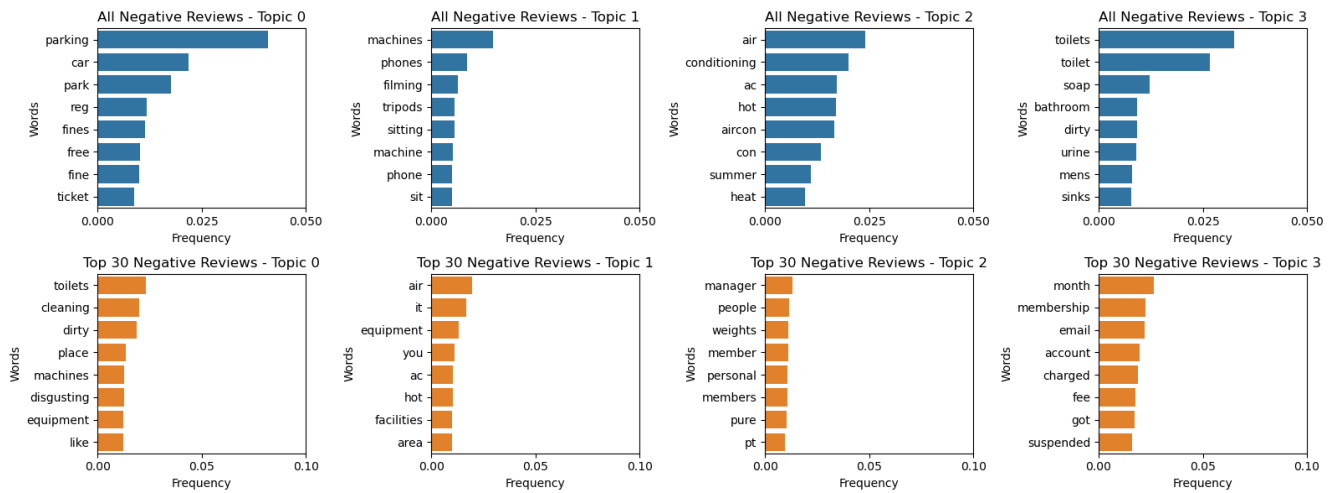


Figure 4: Comparison of Topic Word Frequencies: Top 30 Highest-Volume Negative Review Gyms vs. All Negative Reviews from Same Locations; Blue bars represent frequency words for all negative reviews from common locations combined. Orange bars represent frequency words for the locations with the to 30 highest numbers of negative reviews

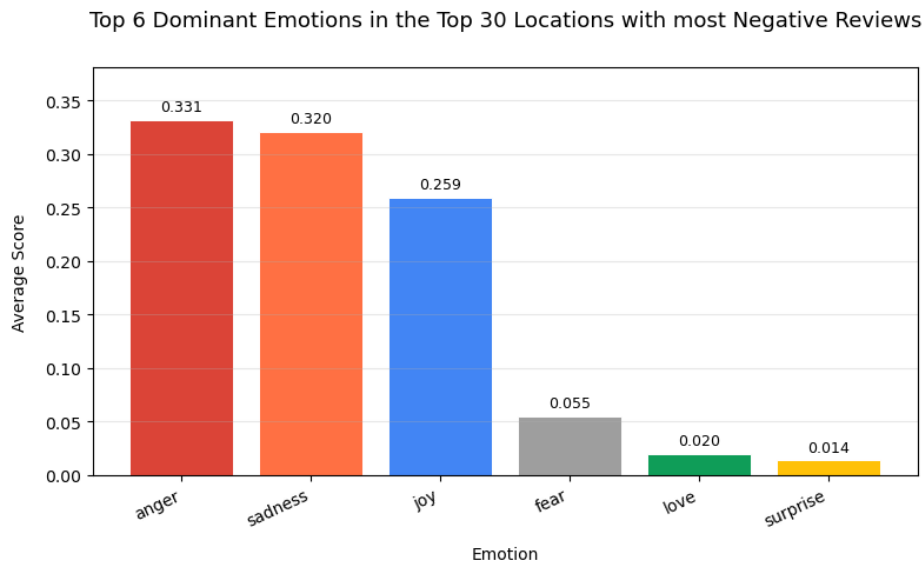


Figure 5: Distribution of emotional categories in negative customer reviews for the 30 PureGym locations with the highest complaint volumes, as classified by a BERT-based emotion model. Data from both Google Reviews and Trustpilot are combined to illustrate the prevalence of specific negative emotions driving customer dissatisfaction across the UK network.

Comparison of emotion distribution between Google Reviews and Trustpilot revealed consistent patterns across platforms, with anger and sadness dominating

negative reviews (Figure 6). This alignment indicates that customer dissatisfaction reflects systemic operational issues rather than platform-specific biases, validating the robustness of our sentiment analysis methodology. However, joy consistently appeared in negative reviews from both platforms, suggesting potential model misclassification requiring further tuning or data preprocessing. The cross-platform consistency strengthens confidence in identifying core operational issues while highlighting the need for model optimization before implementing targeted interventions based on these emotional insights.

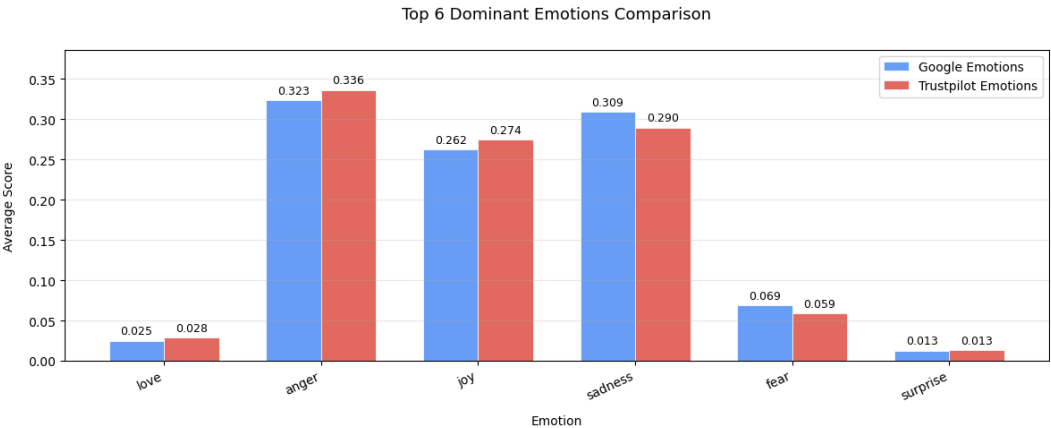


Figure 6: Emotional distribution of negative customer reviews for the 30 PureGym locations with the highest complaint volumes, combining data from Google Reviews and Trustpilot as classified by a BERT-based emotion model.

Finally, we explored the capacity of a small LLM to perform sentiment analysis and generate actionable insights. Table 1 demonstrates how microsoft/Phi-4-mini-instruct synthesizes granular customer concerns into prioritized operational recommendations, transforming topics like “free weights availability” and “hygiene and cleanliness” into concrete business actions. While BERTopic excels at transparent thematic clustering, the LLM adds strategic value by generating context-aware, multi-step recommendations directly from extracted topics. However, it is important to consider that this efficiency comes at the cost of reduced transparency and higher computational requirements.

In this context, the combination of LLM-based insight extraction with traditional topic modeling provides a sensible strategy for both analytical depth and actionable clarity for implementing effective operational changes.

Table 1: LLM-generated insights from negative PureGym customer feedback. The table demonstrates how large language model analysis transforms detailed customer concerns into prioritized, actionable recommendations for operational improvement.

Original Topics	Compressed Insights
free weights availability hygiene and cleanliness reporting feature for issues Air conditioning Pricing Music and advertisements	1. Increase free weights inventory to meet customer demand. 2. Implement regular cleaning schedules to maintain hygiene and cleanliness. 3. Develop a user-friendly reporting feature for members to report issues. 1. Optimize air conditioning costs by implementing energy-efficient systems and scheduling maintenance to reduce energy consumption and expenses. 2. Review and adjust pricing strategy to ensure competitiveness, value for money, and alignment with market trends and customer expectations. 3. Enhance customer experience by curating a diverse and engaging music and advertisement selection, and consider feedback for continuous improvement.
Temperature control Shower temperature User comfort	1. Implement a smart temperature control system to optimize energy usage and enhance user comfort. 2. Regularly monitor and adjust shower temperatures to ensure consistent user satisfaction. 3. Conduct user surveys to identify comfort preferences and adjust facilities accordingly. 4. Invest in high-quality, energy-efficient shower systems to improve user experience and reduce costs. 5. Train staff to understand and maintain optimal temperature settings for maximum comfort and efficiency.
Space and Capacity Accessibility and Facilities Crowd Management and Peak Times Womens area equipment Leg machines Facility layout	1. Optimize gym space utilization to maximize member capacity and revenue. 2. Enhance accessibility features to attract a diverse clientele and improve user experience. 3. Implement crowd management strategies to ensure smooth operations during peak hours. 1. Invest in high-quality, diverse equipment for the women's area to attract and retain female members. 2. Prioritize the addition of leg-focused machines to cater to members' fitness goals and enhance workout variety. 3. Redesign the facility layout to optimize space, improve member flow, and create a more welcoming environment.

5 Conclusions

- Both Google Reviews and Trustpilot revealed identical patterns in negative feedback themes and emotion distribution, demonstrating robust methodology and confirming that operational issues are systemic rather than platform-specific.
- BERT-based emotion analysis effectively identified anger and sadness as dominant emotions in negative reviews, though occasional misclassification (detecting joy in negative content) indicates need for additional fine-tuning to optimize accuracy.
- BERTopic successfully clustered negative reviews around key operational issues, providing transparent and interpretable thematic groupings with lower computational requirements, making it ideal for rapid analysis and operational monitoring.
- LLMs excel at synthesizing complex topics into actionable business recommendations with strategic clarity, however they sacrifice transparency and require higher computational resources compared to traditional models like LDA, which are simpler and faster but need manual interpretation.
- Overall, combining LLM-driven insight extraction with traditional topic modeling and BERT-based sentiment analysis delivers both analytical depth and actionable clarity, enabling efficient translation of customer feedback into operational improvements.

6 References

1. Notebook 1: https://github.com/rgrandym/data_science_cambridge/tree/main/course/reports/course_3#:~:text=Grandy_Rodrigo_CAM_C301_Week_4and5_Topic_project_1.ipynb
2. Notebook 2: https://github.com/rgrandym/data_science_cambridge/tree/main/course/reports/course_3#:~:text=Grandy_Rodrigo_CAM_C301_Week_4and5_Topic_project_2.ipynb